Radar shadow detection in synthetic aperture radar images using digital elevation model and projections

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Abstract. Synthetic aperture radar (SAR) images are currently widely used in target recognition tasks. In this work, we propose an automatic approach for radar shadow detection and extraction from SAR images utilizing geometric projections along with the digital elevation model (DEM), which corresponds to the given georeferenced SAR image. First, the DEM is rotated into the radar geometry, so that each row would match that of a radar line of sight. Next, we extract the shadow regions by processing row by row until the image is covered fully. We test the proposed shadow detection approach on different DEMs and simulated one-dimensional signals and two-dimensional hills and valleys modeled by various variance-based Gaussian functions. Experimental results indicate that the proposed algorithm produces good results in detecting shadows in SAR images with high resolution. © *2014 Society of Photo-Optical Instrumentation Engineers (SPIE)* [DOI: 10.1117/1.JRS.8.083628]

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1 Introduction

Synthetic aperture radar (SAR) contains a large amount of information that offers excellent performance compared to other satellite imaging systems. Despite the advantages, SAR images have shortcomings, such as the relief effect due to the use of imaging radar, which illuminates the ground from oblique views.[1] Thus, the image produced by such radars contains spatial distortions related to geometric characteristics, which are inherent in the formed acquisition geometry.[1,2] Some of these geometric distortions can be reversed, for example in the case of "foreshortening."[1] If the terrain model is available *a priori*, it can be removed using the geocoding process.

There have been efforts to recover or identify shadows in real and very high-resolution images,[3,4] and for shape from shadow,[5] moving cast shadows,[6] and determining building heights[7] applications. Unfortunately, in the case of shadows present in an SAR, such remedies are not possible due to an irreversible lose of information. In fact, the radar will not be able to recover hidden areas, which are the result of a lack of information concerning those particular spatial regions. Shadow regions in SAR imagery are unique in the sense that they can be characterized by the absence of distortion as well as the absence of typical speckle noise, which affects other parts of the images. Nevertheless, a dark shadow area present in an SAR image can either depict a smooth region corresponding to flat or bare soil or water surfaces. This induces a confusion in inferring these regions.[8,9] Moreover, typically, the differences between these areas are huge with the same radar image representation. To detect shadows in SAR images, previous works rely on the characteristics of shadow areas, such as the absence of speckles[2] or distortion free areas. These were complicated and time-consuming procedures since detecting all the areas not affected by at least one such distortion type and locating areas which necessarily contain shadows can be cumbersome. Hence, we require a method which can detect shadows with little

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computational overhead and takes into consideration the complicated nature of the SAR image acquisition process.

In this paper, we propose a simple approach based on one dimensional (1-D) geometrical projections using the digital elevation model (DEM) corresponding to the georeferenced SAR image. We first rotate and fit the DEM to the radar geometry, so that each row would be in the line of sight of the radar. Then, the shadow extraction is done row by row according to the image size, thus simplifying the process to 1-D space. The proposed approach can provide a better target classification and a precise height estimation by using only the shadow information in high-resolution SAR images. A preliminary version appeared in Ref. 10 in which some preliminary example results are given. Here, we elaborate further and synthetic examples in 1-D and 2-D are given to highlight the shadow detection. Moreover, the scheme is illustrated with multiple SAR/DEM imagery and the pipeline is automatic with an easy to use methodology.

The rest of the paper is presented as follows. In Sec. 2, we start with the characteristics of SAR imagery and then we illustrate our methodology for shadow detection. Experimental results are given next in Sec. 3. Finally, Sec. 4 concludes the paper.

2 Shadow Characterization and Detection in Synthetic Aperture Radar Images

Detecting radar shadows includes different characterizations. Several different definitions give rise to different detection methods.[2] Here, the provided factors, which affect the shading, are introduced and a modeling mechanism is also introduced. The incidence angle (Fig. 1) is defined as the angular shift from the vertical position of the sensor viewing in the direction of the target. It is an important parameter which manages the shape of the shadow. For a fixed distance from the nadir, the shadow is an increasing function of the incidence angle. If we increase the sensor height with respect to a given target, then the shadow regions shrink.[11] However, the resolution of the target gets reduced as we move away from the target. On the other hand, for a fixed altitude of the sensor, there are other parameters which can affect the amount of shadow regions; for example, the target altitude can influence the shadow regions. If the slope angle of the overall terrain is higher than the depressing angle, then the shaded areas return very weak responses to the sensor. Moreover, the regions with high relief are due to a low-depression angle and are a source of large shadows present. Thus, we see that the loss of information due to shadows is a decreasing function of the depression angle. Radar shadows produce a 3-D visualization effect without using a stereoscope, but the information provided by the shadow effect on the altitude is not reliable. Note that a radar image superimposed with shadows does not show significant elevations close to nadir. Moreover, the shadows closer to nadir are not detected at all.

Based on the above observations and the fact that SAR recovers a given scene line by line, we propose the following steps for effective identification of shadow regions.

1. The given SAR image is first georeferenced. For this purpose, we utilize the BEAM (Basic ENVISAT Toolbox for (A)ATSR and MERIS) software distributed by the

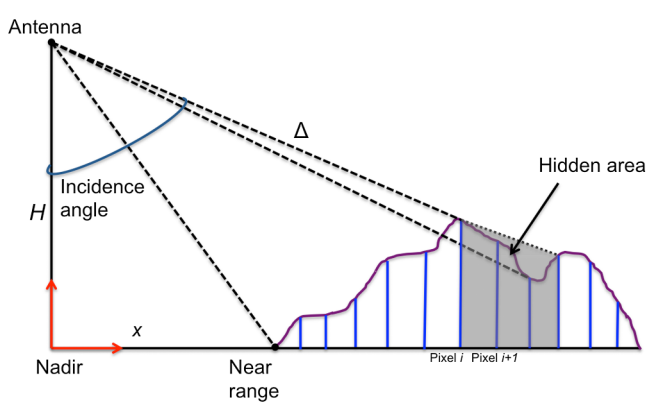


Fig. 1 Illustration of shadow phenomenon on a given row of SAR image.

European Space Agency (ESA) for SAR image georeferencing as we tested the developed approach primarily on ASAR ENVISAT data.

2. Using the geoprojected SAR image, BEAM provides the corresponding incidence angle image [see Fig. 2(f)].
3. The DEM image is rotated, so that each row would match with that of the radar line of sight (slant range).[12] For validation, we use a 30 m resolution ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer) DEM, http://asterweb.jpl.nasa.gov/gdem.asp downloaded from the USGS (United States Geological Survey) website http://gdex.cr.usgs.gov/gdex/, see Fig. 2 for example.
4. The shadow mask is generated from the rotated DEM using the 1-D geometrical projections applied to each row of the DEM. That is, if we let H be the sensor height, then the projection line is given by

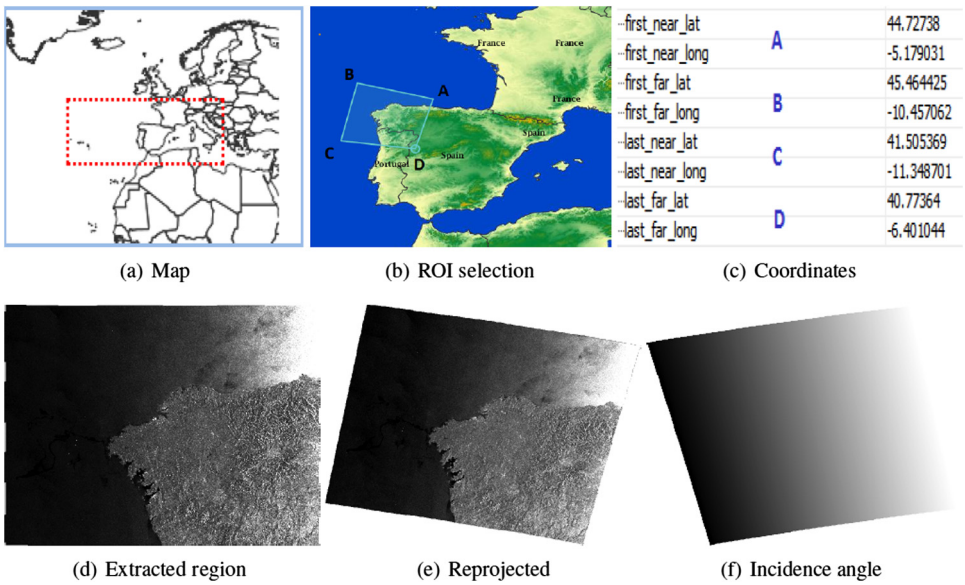


Fig. 2 Illustration of the region/DEM region of interest (ROI) extraction from the north of Portugal and Spain from USGS. (a) The region is shown in a global map for illustration purpose, (b) the rectangular selection region with ABCD indicating the corner points is selected, (c) co-ordinates of the four corner regions, (d) extracted ROI, (e) reprojected/georeferenced image, (f) incidence angle map.

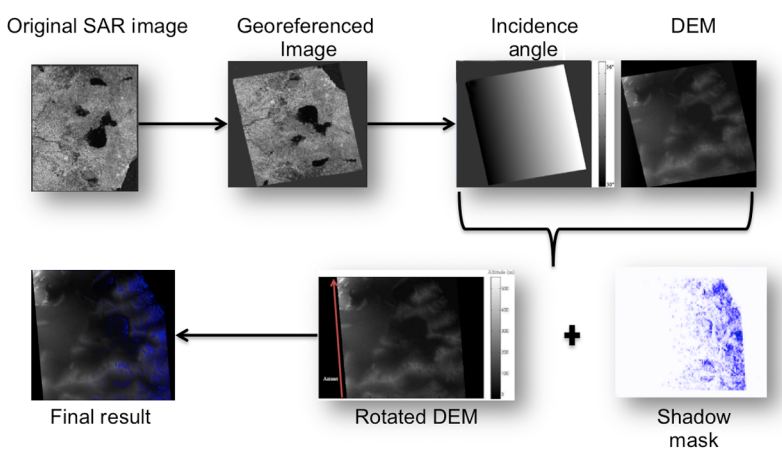


Fig. 3 Shadow detection algorithm. The flowchart corresponds to the DEM-based 1-D projections scheme, and final result shows the shadow regions in blue color. The azimuth is shown as a red arrow in the rotated DEM image. For color referred in this figure see the online PDF version.

$$l(x) = a(x) \cdot x + H, \tag{1}$$

where $a(x)$ represents the slope value computed at each pixel x in the line of sight, see Fig. 1. Note that H is the tunable parameter indicating the height where the antenna is present. Then, a superposition of the DEM and the shadow mask is done, see Fig. 3 for the blue color regions.

Thus, the methodology we proposed is composed of two major parts. First, the shadow detection on the DEM profile is done in a row-by-row fashion. Second, the DEM and shadow mask are merged together to obtain the result. Figure 3 summarizes the proposed approach. The flowchart given in Fig. 3 explains all the five steps previously mentioned, starting with a given SAR

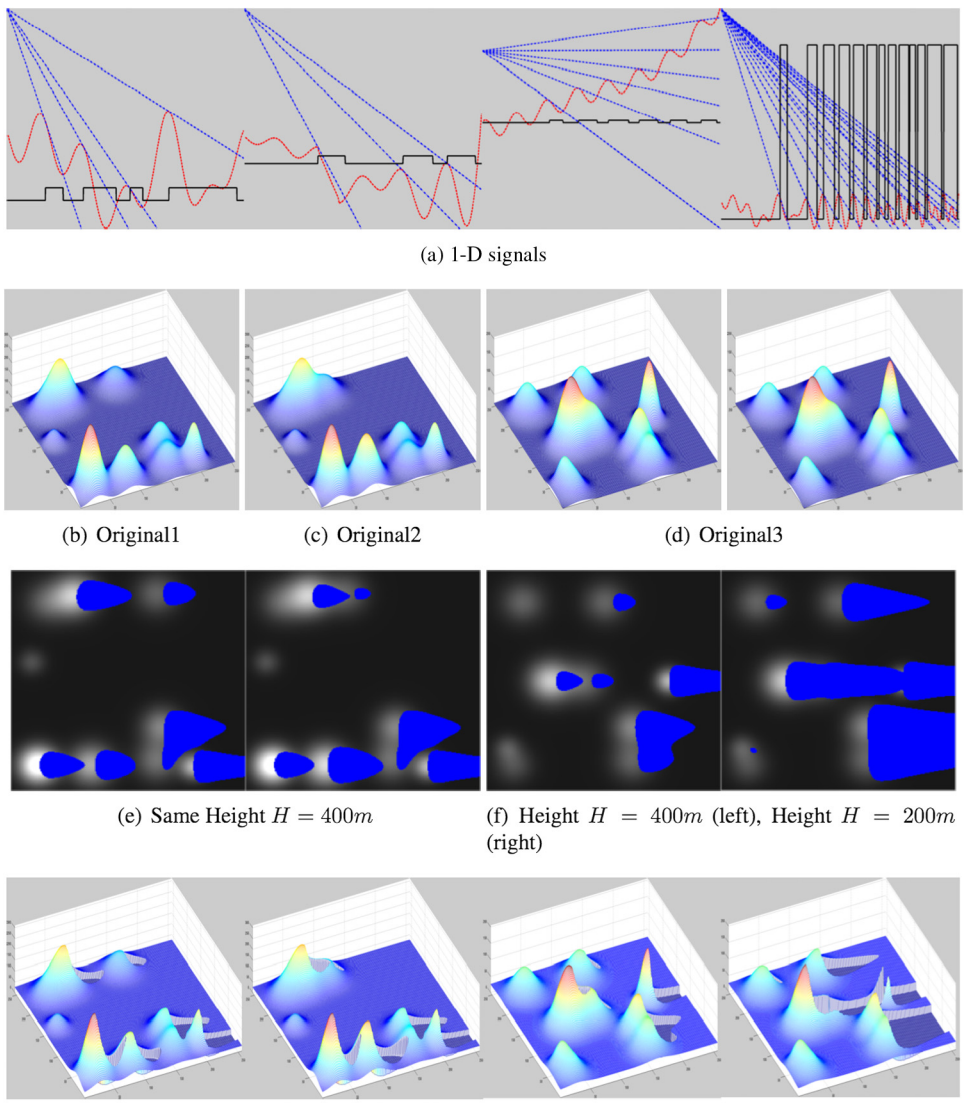


Fig. 4 Synthetic test images illustrating shadow detection method. (a) 1-D signals generated using combinations of sinusoidal, logarithmic, and Gaussian functions. The blue (dash-dotted) lines are the 1-D projections, the red (dashed line) is the DEM or the terrain and the black solid lines are the detected shadows. We use multiple 2-D Gaussian functions modeling various relief scenarios in surface visualization (b, c) different mountains + same Radar Height, (d) same mountains + different radar heights. (e, f) corresponding detected shadow regions for (b)–(d), (g) corresponding surface visualization with shadows superimposed on the terrain. (Video 1, MOV, 0.4 MB [URL: http://dx.doi.org/10.1117/1.JRS.8.083628.1]; Video 2, MOV, 0.5 MB) [URL: http://dx.doi.org/10.1117/1.JRS.8.083628.2].

image with corresponding DEM. We note that in our study, the data elevation model is presented in a grayscale image from zero meter (black) to 250 m (white). The incidence angle is also presented in a grayscale image from 30 (black) to 36 deg (white).

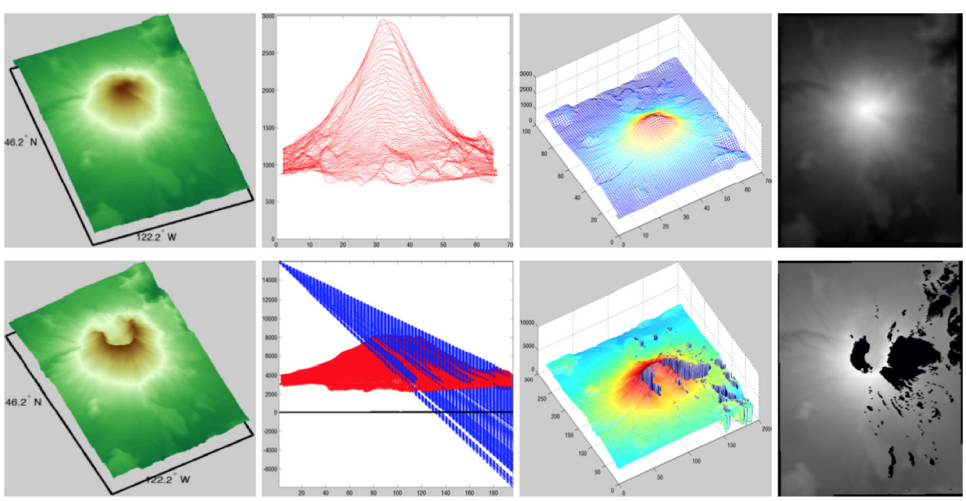
3 Experimental Results

To test the robustness of the proposed shadow detection approach, we perform some simulated profiles corresponding to different 1-D signals using combinations of sinusoidal, logarithmic, and Gaussian functions. The final detection results shown in Fig. 4(a) clearly indicate shadow regions detected, as it consists of finding the intersection of the radar line with the elevated profile from the DEM. Here, the shadow regions are indicated as 1-D square signals (a shadow mask,[13] in black color), in which the zero (flat) values indicate the absence of shadows and nonzero (bumps) values indicate detected shadow intervals. We next use multiple 2-D Gaussian functions to simulate different relief levels in an image. To reiterate our approach, we consider the following steps:

(1) Divide the initial model line by line.
(2) Shadow detection for each line of sight using projections along the lines.
(3) Superposition of the shadow mask on the Gaussian DEM.

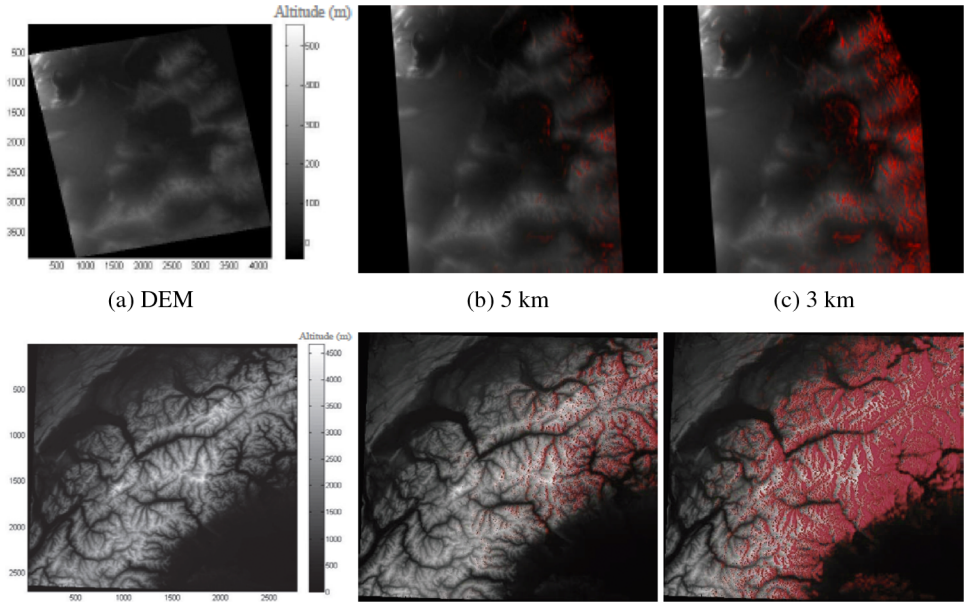
Figures 4(b)–4(d) show that the synthetic 2-D Gaussian functions simulating various scenarios and corresponding shadow regions detected by our approach are provided in 2-D image regions [blue, Figs. 4(e)–4(h)] and in surface visualization [Fig. 4(g)]. It is clear that we do not detect shadows near the nadir. Nevertheless, the detected shadow region's shape provides a confirmation of variations in the relief. Figure 5 shows two real examples for the shadow detection algorithm with respect to different DEMs [Fig. 5(a)] of smaller dimensions and corresponding results with the same radar height of 16 km [Figs. 5(c) and 5(d)]. This illustrates the robustness of the 1-D projections [Fig. 5(b)] based scheme.

Next in Fig. 6, we consider SAR images and their corresponding DEMs to test our algorithm further. Figures 6(a) and 6(d) show two different DEM regions with different characteristics. The final shadow detection results for different radar heights are given in Figs. 6(b), 6(c), 6(e), and 6(f), respectively. As can be seen, the scheme provides reliable shadow estimation across the variations in reliefs.



(a) DEM (b) Projections (c) Shadows (d) 2-D

Fig. 5 Example DEM areas and corresponding shadow detection results. (a) DEM, (b) 1-D projections superimposed on the contour maps of DEM, (c) detected shadows using the DEM, (d) detected shadows projected onto the 2-D image for visualization. First row is the no shadows case (size of DEM 94×66) and the second row is the shadows present case (size of DEM 280×196).



(a) DEM (b) 5 km (c) 3 km

(d) DEM (e) 150 km (f) 3 km

Fig. 6 Shadow detection results for two regions with variation in reliefs and with different altitudes. Top row: Tunisia. Bottom row: Alps mountains. Different altitudes with detected shadow regions are given.

Further visualizations of the detected shadow regions and corresponding DEMs are available online http://dx.doi.org/10.6084/m9.figshare.659896, in which MATLAB figure files showing 3-D maps visualizations and data files (.fig, .mat) are provided as well (See http://sites.google.com/site/suryaiit/research/shadows). Code is available upon request. We believe that the approach presented here can augment a colored remote sensed data-based approach[14] for the recognition of backscattering objects on SAR images via their shadows.

4 Conclusion

In radar imaging, the shadows can give additional information, which can be used for target recognition. For example, the target height can be obtained by the geometry of observation system and the length of shadows, assuming either level or known ground.[14,15] In this paper, a simple solution is given to identify the shadow regions using DEM and 1-D projections. Experimental results with synthetic and real DEMs indicate the robustness of our approach in obtaining shadow regions.

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